

FANUC Robot **series** **GEN II and GEN III**

OVERSLUNG RAIL INSTALLATION MANUAL

Use with the Gen II and Gen III Over Slung Rail Operator's Manual

MARXCG2OR07151E REV B

ORIGINAL INSTRUCTIONS

Thank you very much for purchasing a FANUC robot.

Before using the robot, be sure to read the, *FANUC Robot SAFETY HANDBOOK (B-80687EN)* and understand its contents.

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In this manual, we endeavor to include all pertinent matters. There are, however, a very large number of operations that must not or cannot be performed. Please assume that any operations that are not explicitly described as being possible are not possible.

SAFETY

Safety Precautions

This chapter must be read before using the robot. For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell. For safe use of FANUC robots, you must read and follow the instructions in the *FANUC Robot SAFETY HANDBOOK (B-80687EN)*.

Definition of User

Personnel can be classified as follows:

Operator • Turns the robot controller power on/off • Starts the robot program from operator panel
Programming or Teaching Operator • Operates the robot • Teaches the robot inside the safeguarded space Maintenance engineer
Maintenance Technician • Operates the robot • Teaches the robot inside the safeguarded space • Maintenance (repair, adjustment, replacement)

The operator is not allowed to work in the safeguarded space.

The programmer or teaching operator and maintenance engineer are allowed to work in the safeguarded space. Work carried out in the safeguarded space include transportation, installation, teaching, adjustment, and maintenance.

To work inside the safeguarded space, the person must be trained on proper robot operation.

Table s-1 lists the work outside the safeguarded space. In this table, the symbol “○” means the work is allowed to be carried out by the worker.

Table s-1 Work Performed Outside the Safeguarded Space

	Operator	Programmer or Teaching Operator	Maintenance Technician
Turn power ON/OFF to Robot controller	○	○	○
Select operating mode (AUTO, T1, T2)		○	○
Select remote/local mode		○	○
Select robot program with teach pendant		○	○

Select robot program with external device		○	○
Start robot program with operator's panel	○	○	○
Start robot program with teach pendant		○	○
Reset alarm with operator's panel		○	○
Reset alarm with teach pendant			
Set data on teach pendant			
Teaching with teach pendant			
Emergency stop with operator's panel	○	○	○
Emergency stop with teach pendant	○	○	○
Operator Panel maintenance		○	
Teach Pendant maintenance			○

During robot operation, programming and maintenance, the operator, programmer, teaching operator and maintenance engineer take care of their own safety using at least the following safety protectors:

- Use clothes, uniform, overall adequate for the work
- Safety shoes
- Helmet

Definition of Safety Notations

To ensure the safety of workers and prevent damage to the machine, this manual indicates each precaution on safety according to its severity identified below. Read the contents of each **Warning** and **Caution** before attempting to use the robot.

 WARNING
Indicates a hazard resulting in the death or serious injury of the user could occur if he or she fails to follow the approved procedure.
 CAUTION
Indicates a hazard resulting in the injury of the user or damage to equipment could occur if the user fails to follow the approved procedure.

Note

Indicates a supplementary explanation not related to any WARNING or CAUTION.

Safety Precautions for Paint Robots

Paint Robot Safety Overview

Today's highly automated paint booths require that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between any moving machinery along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment is usually provided by the paint shop supervisor.

CAUTION

Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic or Production mode
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is operated at full production speed. It is initiated at the System Control Console (SCC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth whenever automatic mode is initiated.

Paint System Safety Features

Process technicians and maintenance personnel must become familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features (in addition to other standard robot safety features):

- Some paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.
- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.

- An intrinsically safe teach pendant is used when personnel are inside the booth while paint is being atomized.
- Additional safety components for Paint systems such as Light curtains, door switches etc. are defined by a Risk Assessment.
- System safety devices are monitored by hardware circuits and/or a safety PLC. The robots are interlocked to the hardware circuits and/or safety PLC.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

WARNING

Never bypass, strap, or otherwise deactivate the robot purge system for any reason. Deactivating the purge system might result in an explosion serious injury or death.

WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station.
- Know the work envelope of the robot and hood/deck and door opening devices.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Be aware of signals and/or operations that could start up paint process application equipment such as guns or bells.
- Be aware of potential air pressure buildup behind paint robot access covers.
- Be aware of all safety precautions when dispensing of paint is required.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust cannot enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust.
- Do not energize the robot until you have performed the following steps.
 1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.

2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
4. When the dust has been removed, the covers must be replaced immediately.
5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

Grounding: All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.

High Voltage: High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.

1. Avoid any accumulation of combustible vapors or coating matter.
2. Follow all manufacturer recommended cleaning procedures.
3. Make sure all interlocks are operational.
4. No smoking.
5. Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
6. Disable all air and paint pressure to the paint applicator.
7. Verify that the lines are not under pressure.
8. Be aware of potential hazards presented by paint process hardware, such as rotating or spinning components on the paint applicator.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

1. Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
2. Be sure all covers and inspection plates are in good repair and in place.
3. Whenever possible, turn off the main electrical disconnect before you clean the robot.
4. When using solvents, observe the following:
 - Wear eye protection and protective gloves during application and removal.

- Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
- If there is contact with the skin, wash with water.
- Follow the Original Equipment Manufacturer's Material Safety Data Sheets.

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1 OVERVIEW

The software options, R831, iRail Motion for Gen II OverSlung Rail and R894, iRail Gen. III for Gen III OverSlung Rail, support the following features:

- Provide robot library setup for *i*Rail unit as an extended axis or a separate motion group.
- Provide current gravity torque data for *i*Rail unit status display.
- Support runtime gravity torque check to maximize the *i*Rail bearing life.
- Revise full reversal CNTxx rail motion to CNT0 internally in motion system.

Note

- Gen II is supported on R-30iA, R-30iB, and R-30iB Plus controller platforms
- Gen III is supported on R-30iB and R-30iB Plus controller platforms

2 iRAIL MOTION CONFIGURATION WITH A G1 EXTENDED AXIS

2.1 iRAIL Motion Configuration with a G1 Extended Axis

About this task

This section describes the procedure to configure the rail unit as an extended axis, and display or modify existing extended axis configuration.

Before you begin

Be sure the following options are installed:

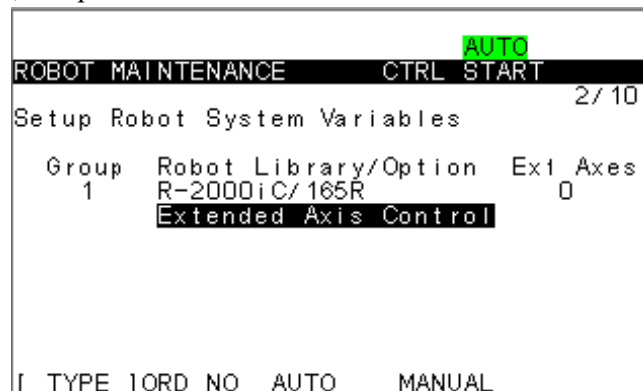
- iRail Motion (R831)
- Extended Axis Control (J518)
- Constant Path (R663)

Procedure

1. At Controlled Start, press the **MENU** key. You will see a screen similar to the following.



2. Select Maintenance, and press ENTER. You will see a screen similar to the following.



3. Select **Extended Axis Control**, and press F4, **MANUAL**.

```
**** EXTENDED AXIS SETTING PROGRAM ****

-- SELECT GROUP --

0. EXIT
1. Group 1
```

4. Select the desired group.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 Initialization ****

-- Hardware start axis setting --
Enter hardware start axis
(Valid range: 1 - 32)
Default value = 7
```

5. Press ENTER to select the default hardware start axis value.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 Initialization ****

*** Group 1 Total Ext Axis = * * *
1. Display/Modify Ext axis 1~3
2. Add Ext axes
3. Delete Ext axes
4. EXIT
Select? █
```

6. Select Add Ext axes.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 Initialization ****

*** Group 1 Total Ext Axis = * * *
1. Display/Modify Ext axis 1~3
2. Add Ext axes
3. Delete Ext axes
4. EXIT
Select? 2
Enter axis to add (1~3)?
1█
```

7. Enter 1 to add axis 1.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- EXTENDED AXIS TYPE --

1. Integrated Rail (Linear axis)
2. Integrated Arm (Rotary axis)
3. Auxiliary Linear Axis
4. Auxiliary Rotary Axis
5. OverSlung Rail Gen II (Integrated)
6. OverSlung Rail Gen II (Auxiliary)

Select? █
```

8. Select **OverSlung Rail Gen II (Integrated)** for integrated extended axis.

```

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- Rail Length (mm) --

  1.  4500      6. 13500      11. 21000
  2.  6000      7. 15000      12. 22500
  3.  9000      8. 16500      13. 24000
  4. 10500      9. 18000      14. 25500
  5. 12000     10. 19500      15. 27000
  0. Next page
Select? █

```

9. Select the desired rail length, e.g. enter 5 for 12000 mm.

```

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- T-Car Option --

T-Car Option = None
Enter (1:None, 2:LH, 3:RH, 4:LH&RH)? █

```

10. Select the desired T-Car Option, e.g. enter 4 for LH&RH.

```

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- LH T-Car Payload (kg) --

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- RH T-Car Payload (kg) --

```

11. T-Car payload is not being used currently; enter 0 for both LH and RH T-Car.

```

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- Rail Mastering Position (mm) --

Enter (1:Standard, 2:User definable)? █

```

12. Select **Standard** for standard rail mastering position. For iRail Gen III only, **Standard** requires the length of first rail segment to determine mastering.

```

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- 1st Rail Segment Length (mm) --
Enter length of left-most rail segment
as seen from the front of the robot:
  1.  4500 (master pos = 3500)
  2.  6000 (master pos = 3000)
  3.  9000 (master pos = 3000)
  4. 10500 (master pos = 3500)
  5. 12000 (master pos = 3000)
Select? █

```

If **User definable** is selected, you will see a screen similar to the following.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- Rail Mastering Position (mm) --
Enter Pos (divisible by 1500)? █
```

13. Enter rail mastering position divisible by 1500, e.g. 3000.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- SELECT AMP NUMBER --
Enter amplifier number (1~84)? █
```

14. Enter desired amplifier number.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- SELECT AMP TYPE --

1. A06B-6400 series 6 axes amplifier
2. A06B-6240 series Alpha i amp. or
   A06B-6160 series Beta i amp.

Select? █
```

15. Select the Alpha or Beta amp.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- BRAKE SETTING --
Enter Brake Number (0~32)? █
```

16. Enter the desired Brake Number.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- SERVO TIMEOUT --
Servo Off is Disable
Enter (1: Enable 2: Disable)? █
```

17. Select **Enable** Servo Timeout.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

-- SERVO TIMEOUT VALUE --
Enter Servo Off Time? (0.0~30.0 Sec) █
```

18. Enter the servo off time value.

2.2 Display or Modify the Existing Extended Axis Configuration

Procedure

1.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 Initialization ****

*** Group 1 Total Ext Axis = 1 * *
  1. Display/Modify Ext axis 1~3
  2. Add Ext axes
  3. Delete Ext axes
  4. EXIT
Select? 1
Enter Axis to Display/Modify(1~3)?
1
```

2. Select “Display/Modify Ext axis 1~3”, then enter 1 for extended axis 1 configuration.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

  1. Integrated Rail (Linear axis)
  2. Integrated Arm (Rotary axis)
  3. Auxiliary Linear Axis
  4. Auxiliary Rotary Axis
  5. OverSlung Rail Gen II (Integrated)
  6. OverSlung Rail Gen II (Auxiliary)

Select?
Default value = 5
```

3. Press ENTER to select default value.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

  1. Motor Type = aiS40/1000 160A
  2. Gear Ratio = 50.000
  3. Ext Type = Integrated Rail
    Ext_direction = Y
  4. Joint Max Speed = 3000.000(mn/s)
Press ENTER to Continue
```

4. Press ENTER to continue.

```
**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

  5. Motion Sign = FALSE
  6. Upper Limit = 8900.000mm
  7. Lower Limit = 1300.000mm
  8. Master Position = 3000.000mm
  9. Accel Time1 = 1360
    Accel Time2 = 680
Press ENTER to Continue
```

5. Press ENTER to continue.

```

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

11. Minimum Accel Time = 256
12. Load Ratio = 5.000
13. Axis amp number = 2
14. Axis amp type = 2
15. Brake Number = 1
16. Servo Off Enable = Enable
17. Servo Off Time = 25.000 sec
Select (11~17: Modify, 0: Next Items)?

```

6. Select 0 for next items.

```

**** EXTENDED AXIS SETTING PROGRAM ****
**** Ext Axis G: 1 A: 1 Initialization *

18. Rail Length = 12000.000mm
19. T-Car Option = LH & RH
20. Rail Master Pos = 3000.000mm
Select (18~20: Modify, 0: Exit)?

```

7. Select 18~20 to modify existing configuration, or 0 to exit the display menu.

```

                                AUTO
ROBOT MAINTENANCE      CTRL START
                        FUNCTIONS
1  R-200  1 START (COLD) / 10
   Exten  2 RESTORE/BACKUP
          3 PRINT SCREEN
          4 UNSIM ALL I/O

[ TYPE ]ORD NO  AUTC  MANUAL

```

8. When the configuration is done, exit the extended axis setup menu, then press Fctn key and select START (COLD) to complete the setup.

3 iRAIL MOTION CONFIGURATION WITH G2 RAIL UNIT

About this task

This section describes the procedure to configure the rail unit as a separate motion group.

Before you begin

Be sure the following options are installed:

- iRail Motion (R831)
- Rail Unit (H894) for G2 Robot Model
- Constant Path (R663)

Procedure

1. At Init Start, you will see a screen similar to the following.

```
Is Rail Type OverSlung Rail Gen II ?  
Enter (1: TRUE, 2: FALSE)? █
```

2. Select **TRUE**, and press ENTER.

```
-- Robot Mounted On The Rail Setting --  
Enter Robot Group Number ? █
```

3. Enter the group number of the robot that is mounted on the rail unit.

```
-- Rail Length (mm) --  
1. 4500    6. 13500   11. 21000  
2. 6000    7. 15000   12. 22500  
3. 9000    8. 16500   13. 24000  
4. 10500   9. 18000   14. 25500  
5. 12000   10. 19500  15. 27000  
0. Next page  
Select? █
```

4. Select desired rail length. The subsequent configuration will be same as Extended Axis setup.

4 GRAVITY MOMENT LIMIT

This section describes the gravity moment limit and the alarm IRAL-001 Exceed Gravity Torque. During teaching mode, gravity moment percent will be displayed on the upper right corner of the *i*Pendant.

Gravity moment percent =

Current gravity moment

Maximum gravity moment allowed

Gravity moment percent

L66%

100%

Busy

Stop

Hold

Fault

Run

I/O

Prod

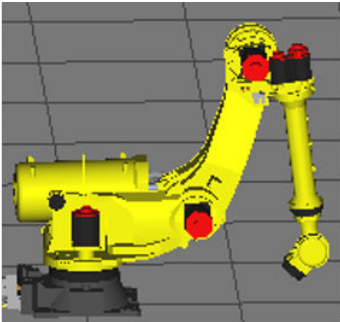
TCyc

TEST1A LINE 0

T2

ABORTED

JOINT



When the robot arm is tucked in, the gravity moment percent will be less than 100%.

L113%

100%

Busy

Stop

Hold

Fault

Run

I/O

Prod

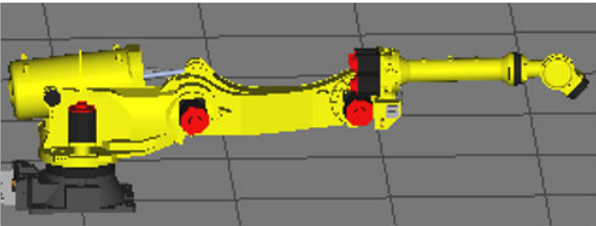
TCyc

TEST1A LINE 0

T2

ABORTED

JOINT



When the robot arm is extended out, the gravity moment percent will be greater than 100%.

Alarm IRAL-001

IRAL-001 Exceed Gravity Torque

L103%

10%

Busy

Stop

Hold

Fault

Run

I/O

Prod

TCyc

T2

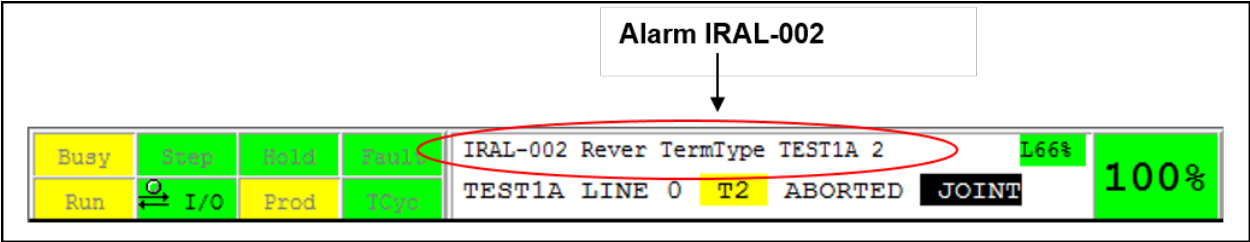
G1 S JOINT

5 REVERSAL CNTXX RAIL MOTION

This section describes the reversal rail motion and the alarm IRAL-002 Rever TermType ProgName Line#.

The software will automatically detect and reduce potentially high acceleration during rail reverse motion.

The alarm IRAL-002 Rever TermType ProgName Line# will be posted if potentially high acceleration is detected during rail reverse motion. IRAL-002 is a warning alarm, no TP program change is required, and software will automatically reduce the aggressive acceleration internally during the rail reverse motion.



For example, during the execution of the following TP program, system will post the warning alarm IRAL-002 when P[1] to P[2] is a long rail motion segment with CNT100 term type at P[2]. There is no need to change the TP program, software will automatically detect and reduce the aggressive acceleration during the rail reverse motion.

```
TEST1A
1:J @P[1] 100% FINE
2:J P[2] 100% CNT100
3:J @P[1] 100% FINE
```


REVISION RECORD

Edition	Date	Contents
A	7/2015	• First revision.
B	january, 2020	• Included information to support GEN III Overslung Rails.

